

National Neighborhood Data Archive (NaNDA): Liquor, Tobacco, and Convenience Stores by Census Tract, United States, 2003-2017



openICPSR-123541

nanda_lqtbcon_tract_2003-2017_01P.dta

nanda_lqtbcon_tract_2003-2017_01P.csv

nanda_lqtbcon0317T_01P.sas7bdat

Overview and Data Dictionary

Documentation Version: 1.1

Last updated: 12/7/2020

Dataset Overview

Description

This dataset contains measures of the number and density of liquor, tobacco, and convenience stores per United States census tract from 2003 through 2017.

Principal Investigators

- Jessica Finlay, University of Michigan Institute for Social Research
- Mao Li, University of Michigan Institute for Social Research
- Michael Esposito, University of Michigan Institute for Social Research
- Iris Gomez-Lopez, University of Michigan Institute for Social Research
- Anam Khan, University of Michigan Institute for Social Research
- Philippa Clarke, University of Michigan Institute for Social Research
- Megan Chenoweth, University of Michigan Institute for Social Research

Funding Sources

- United States Department of Health and Human Services. Administration for Community Living. National Institute on Disability, Independent Living, and Rehabilitation Research (90RTHF0001)
- United States Department of Health and Human Services. National Institutes of Health. National Institute on Aging (RF1-AG-057540)

Data Sources

The North American Industry Classification System (NAICS) was used to identify organizations that can be considered liquor, tobacco, and convenience stores. NAICS was developed by the United States Office of Management and Budget and is “the standard used by Federal statistical agencies in classifying business establishments for the purpose of collecting, analyzing, and publishing statistical data related to the U.S. business economy” (United States Census Bureau, 2017).

Establishment data was taken from the National Establishment Time Series (NETS) database, which provides annual records of the US economy for more than 60 million private for-profit and nonprofit establishments and government agencies (Walls & Associates, 2017). This longitudinal data source classifies establishments by NAICS code and provides detailed business microdata and address history. NETS is produced in most years by Walls and Associates using archival establishment data from Dun & Bradstreet. While it currently includes data from 1990 to 2015 and 2017, only data from 2003-2015 and 2017 were used to create this dataset. (NETS data was not produced for 2016.)

Census tracts for establishments were determined using the 2010 TIGER/Line shapefiles produced by the United States Census Bureau (United States Census Bureau, 2010).

Census tract population figures for 2000-2010 were taken from Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010 (Clarke & Melendez, 2019). Population figures for 2011-2017 were obtained from Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017 (Melendez et al., 2020).

Coverage

The dataset contains one observation per census tract per year in the United States, Puerto Rico, and the US Virgin Islands. Other US island territories are excluded.

Methodology

This dataset highlights the number and density of liquor, tobacco, and convenience stores in neighborhoods of the United States. Studies have investigated the relationship between these

establishments and domestic violence (Livingston, 2011), crime (Jennings et al., 2014), and overall life expectancy (Galiatsatos et al., 2018). Convenience stores and liquor stores in particular, along with fast food restaurants and dollar stores, are common sources of food in disadvantaged neighborhoods where access to healthy food (such as in supermarkets and farmer's markets) is limited. Prior research has examined the effects of unhealthy food outlet density on self-rated health, healthy eating habits, obesity, and colorectal cancer, among other health outcomes (Xin et al., 2019, Canchola et al., 2017, Spring, 2018, Rummo et al., 2015). While tobacco stores in particular are not a common food destination, they have implications for health outcomes such as smoking (Galiatsatos et al., 2020). Unhealthy food outlets and tobacco stores are unevenly concentrated in impoverished neighborhoods, while access to healthy food outlets such as supermarkets is limited in these areas (Zenk et al., 2005, Larson et al., 2009).

Our research team identified the following NAICS codes as representing liquor, tobacco, and convenience stores:

- 4453: beer, wine, and liquor stores (also known as package stores).
- 453991: cigar, cigarette, and tobacco stores, excluding stores that sell electronic cigarettes.
- 445120: convenience stores without gas stations. These are defined by NAICS as "establishments...primarily engaged in retailing a limited line of goods that generally includes milk, bread, soda, and snacks."
- 447120: gas stations with convenience stores.

For each NAICS code we identified, we queried the NETS database for all establishments that were open at some time between 2003 and 2015 and in 2017 (NETS does not contain data for 2016), and their location during each year. Note that NETS data were not cleaned or verified during this process, and that NETS data has limitations and inaccuracies as described in Gomez-Lopez et al. (2017) and Finlay et al. (2019).

We determined each establishment's census tract for each year by mapping its latitude and longitude provided in the NETS database to the 2010 version of the TIGER/Line shapefiles from the US Census Bureau.

Using these locations, we created three variables to represent establishment counts per tract. The first count variable is the total of all establishments with that NAICS code in the tract. The second is the count of just establishments with non-zero sales figures in the current year, which can be used where desired to filter out establishments with very low business volume. The third is the count of establishments employing more than one person, which can be used to filter out very small owner-operated businesses if desired. The most useful variable for a given research question may vary by establishment type, so we have created three count types for all NAICS codes so that users may make their own decision about the version that best meets their needs.

We then calculated population densities (per 1000 people) for each of the three count variables for each NAICS code. Tract populations for 2003-2009 come from the NaNDA dataset [Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United](#)

[States, 2000-2010](#). Values for each year are interpolated. For 2010-2012 and 2013-2017, we used the tract population from the NaNDA dataset [Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017](#). Population figures in this dataset were taken from the 2012 and 2017 ACS five-year estimates, respectively, in order to avoid overlapping ACS estimates as advised by the Census Bureau, and for compatibility with other NaNDA datasets. Population figures for 2010-2017 are not interpolated. Consult the codebooks for the NaNDA datasets Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010 (Clarke & Melendez, 2019) and Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017 (Melendez et al., 2020) for details on why these population figures were selected and how 2003-2009 population was interpolated.

We also created three variables representing the density of establishments per square mile. The area is the land area of the tract from the 2010 version of the TIGER/Line shapefiles. As a result, all area densities are standardized to 2010 census geography and are comparable across years across the entire dataset.

Population density was set to -1 for tracts with zero population in a given year. Population and area densities are missing for tracts in Puerto Rico and the US Virgin Islands, which are not included in the TIGER/Line shapefiles. Population and area densities are also missing in certain years for a few census tracts whose FIPS codes changed at some point between 2003 and 2017.

Finally, we merged counts and densities of all selected NAICS codes to create an aggregate dataset containing tract population, area, and count and density variables for all liquor, tobacco, and convenience stores per tract in each year.

Usage Notes

The businesses in this dataset were selected to represent locations where people tend to purchase consumable goods that contribute to negative health outcomes: unhealthy food, alcohol, and tobacco products. We recognize that not every purchase from a convenience or liquor store is unhealthy; that the same products can be purchased from other locations such as supermarkets, fast food restaurants, and bars; and that people are constrained by factors besides health when choosing where and when to purchase food, alcohol, and tobacco.

Data users interested in a broader picture of a neighborhood's food and alcohol landscape might find useful data in these other NaNDA datasets:

- [Dollar stores by census tract, United States, 2003-2017](#) (Gomez-Lopez et al., 2020).
- [Eating and drinking places by census tract, United States, 2003-2017](#) (Esposito et al., 2020), which includes fast food restaurants and bars.
- [Grocery stores by census tract, United States, 2003-2017](#) (Finlay et al., 2020).

Variables

For additional information on the definition of each NAICS code, see the Methodology section of this document and the 2017 NAICS website: <https://www.census.gov/cgi-bin/sssd/naics/naicsrch?chart=2017>.

Variable	Type	Obs	Unique	Mean	Min	Max	Label
tract_fips10	string	1036462	74033	.	.	.	Census tract FIPS code (2010 TIGER/Line shapefiles)
year	int	1036462	14	2009.571	2003	2017	Year
population	float	1020881	250271	4208.894	0	65528	Tract population
aland10	float	1022434	72534	48.07265	0	85425.73	Tract land area, square miles
count_4453	int	1036462	31	0.6307207	0	37	# liquor stores
count_sales_4453	int	1036462	30	0.6000934	0	37	# liquor stores w/ sales>0
count_emp_4453	int	1036462	31	0.5670184	0	34	# liquor stores w/ 2+ employees
popden_4453	float	1015188	149822	0.8094143	-1	289507.3	# liquor stores per 1000 people
popden_sales_4453	float	1015178	145146	0.7211539	-1	289507.3	# liquor stores per 1000 people w/ sales>0
popden_emp_4453	float	1015181	140512	0.7915853	-1	289507.3	# liquor stores per 1000 people w/ 2+ employees
aden_4453	float	1017996	62479	0.9800084	0	180.2766	# liquor stores per sq mile
aden_sales_4453	float	1018000	61696	0.9346063	-1	180.2766	# liquor stores per sq mile w/ sales>0
aden_emp_4453	float	1017997	59505	0.868416	-1	180.2766	# liquor stores per sq mile w/ 2+ employees
count_453991	int	1036462	33	0.1505593	0	44	# tobacco stores
count_sales_453991	int	1036462	30	0.1384122	0	38	# tobacco stores w/ sales>0
count_emp_453991	int	1036462	28	0.1310371	0	42	# tobacco stores w/ 2+ employees
popden_453991	float	1015088	55112	0.051483	-1	2684.664	# tobacco stores per 1000 people
popden_sales_453991	float	1015082	52706	0.0482506	-1	2684.664	# tobacco stores per 1000 people w/ sales>0
popden_emp_453991	float	1015080	49319	0.0404483	-1	536.9328	# tobacco stores per 1000 people w/ 2+ employees
aden_453991	float	1017996	21105	0.1909147	0	97.69116	# tobacco stores per sq mile
aden_sales_453991	float	1017996	20699	0.176573	0	97.69116	# tobacco stores per sq mile w/ sales>0
aden_emp_453991	float	1017996	19829	0.1602974	0	66.45511	# tobacco stores per sq mile w/ 2+ employees
count_445120	int	1036462	28	1.113038	0	30	# convenience stores

National Neighborhood Data Archive (NaNDA)
Liquor, Tobacco, and Convenience Stores by Census Tract, United States, 2003-2017

Variable	Type	Obs	Unique	Mean	Min	Max	Label
count_sales_445120	int	1036462	26	1.061673	0	26	# convenience stores w/ sales>0
count_emp_445120	int	1036462	26	1.054061	0	29	# convenience stores w/ 2+ employees
popden_445120	float	1015300	208582	0.6454008	-1	99110.82	# convenience stores per 1000 people
popden_sales_445120	float	1015285	203433	0.6046398	-1	99110.82	# convenience stores per 1000 people w/ sales>0
popden_emp_445120	float	1015293	202891	0.6252539	-1	99110.82	# convenience stores per 1000 people w/ 2+ employees
aden_445120	float	1017996	102157	1.07573	0	230.2062	# convenience stores per sq mile
aden_sales_445120	float	1017996	100552	1.010282	0	230.2062	# convenience stores per sq mile w/ sales>0
aden_emp_445120	float	1017996	99918	1.008286	0	230.2062	# convenience stores per sq mile w/ 2+ employees
count_447110	int	1036462	6	0.003756	0	8	# gas stations w/ convenience stores
count_sales_447110	int	1036462	6	0.0033653	0	8	# gas stations w/ convenience stores w/ sales>0
count_emp_447110	int	1036462	6	0.0035197	0	8	# gas stations w/ convenience stores w/ 2+ employees
popden_447110	float	1015047	1923	0.0009438	0	1.350439	# gas stations w/ convenience stores per 1000 people
popden_sales_447110	float	1015047	1766	0.0008486	0	1.350439	# gas stations w/ convenience stores per 1000 people w/ sales>0
popden_emp_447110	float	1015047	1842	0.0008921	0	1.350439	# gas stations w/ convenience stores per 1000 people w/ 2+ employees
aden_447110	float	1017996	590	0.0025597	0	42.3765	# gas stations w/ convenience stores per sq mile
aden_sales_447110	float	1017996	586	0.0022515	0	42.3765	# gas stations w/ convenience stores per sq mile w/ sales>0
aden_emp_447110	float	1017996	578	0.0024456	0	42.3765	# gas stations w/ convenience stores per sq mile w/ 2+ employees

References

- Canchola, A. J., Shariff-Marco, S., Yang, J., Albright, C., Hertz, A., Park, S. Y., Shvetsov, Y. B., Monroe, K. R., Le Marchand, L., Gomez, S. L., Wilkens, L. R., & Cheng, I. (2017). Association between the neighborhood obesogenic environment and colorectal cancer risk in the Multiethnic Cohort. *Cancer Epidemiology* 50(A), 99-106. <https://doi.org/10.1016/j.canep.2017.08.009>
- Clarke, P. J., & Melendez, R. (2019). National Neighborhood Data Archive (NaNDA): Neighborhood socioeconomic and demographic characteristics of census tracts, United States, 2000-2010 (V1) [Data set]. Inter-university Consortium for Political and Social Research. <https://doi.org/10.3886/E111107V1>
- Esposito, M., Li, M., Finlay, J. M., Gomez-Lopez, I. N., Khan, A. M., Clarke, P. J., and Chenoweth, M. M. (2020). National Neighborhood Data Archive (NaNDA): Eating and drinking places by census tract, United States, 2003-2017 (V2) [Data set]. Inter-university Consortium for Political and Social Research. <https://doi.org/10.3886/E115404V2>
- Finlay, J. M., Esposito, M., Kim, M. H., Gomez-Lopez, I. N., & Clarke, P. J. (2019). Closure of “third places”? Exploring potential consequences for collective health and wellbeing. *Health & Place*, 60, 102225. <https://doi.org/10.1016/j.healthplace.2019.102225>
- Finlay, J. M., Li, M., Esposito, M., Gomez-Lopez, I. N., Khan, A. M., Clarke, P. J., and Chenoweth, M. M. (2020). National Neighborhood Data Archive (NaNDA): Grocery stores by census tract, United States, 2003-2017 (V1) [Data set]. Inter-university Consortium for Political and Social Research. <https://doi.org/10.3886/E123001V1>
- Galiatsatos, P., Kineza, C., Hwang, S., Pietri, J., Brigham, E., Putcha, N., Rand, C.S. Hansel, N. N. & McCormack, M. (2018). Neighbourhood characteristics and health outcomes: evaluating the association between socioeconomic status, tobacco store density and health outcomes in Baltimore City. *Tobacco Control*, 27(e1), e19-E24. <https://doi.org/10.1136/tobaccocontrol-2017-053945>
- Galiatsatos, P., Brigham, E., Krasnoff, R., Rice, J., Van Wyck, L., Sherry, M., Rand, C. S., Hansel, N. N., & McCormack, M. C. (2020). Association between neighborhood socioeconomic status, tobacco store density and smoking status in pregnant women in an urban area. *Preventive Medicine*, 136(106107). <https://doi.org/10.1016/j.ypmed.2020.106107>
- Gomez-Lopez, I. N., Clarke, P. J., Hill, A. B., Romero, D. M., Goodspeed, R., Berrocal, V. J., Vinod Vydiswaran, V.G, & Veinot, T. C. (2017). Using social media to identify sources of healthy food in urban neighborhoods. *Journal of Urban Health*, 94(3), 429-36. <https://doi.org/10.1007/s11524-017-0154-1>
- Gomez-Lopez, I. N., Esposito, M., Li, M., Finlay, J. M., Khan, A. M., Clarke, P. J., & Chenoweth, M. M. (2020). National Neighborhood Data Archive (NaNDA): Dollar stores by census tract, United States, 2003-2017 (V1) [Data set]. Inter-university Consortium for Political and Social Research. <https://doi.org/10.3886/E123802V1>

- Jennings, J. M., Milam, A. J., Greiner, A., Furr-Holden, C. D., Curriero, F. C., & Thornton, R. J. (2014). Neighborhood alcohol outlets and the association with violent crime in one mid-Atlantic City: the implications for zoning policy. *Journal of Urban Health: Bulletin of the New York Academy of Medicine*, 91(1), 62–71. <https://doi.org/10.1007/s11524-013-9821-z>
- Larson, N. I., Story, M. T., Nelson, M. C. (2009). Neighborhood environments: disparities in access to healthy foods in the U.S. *American Journal of Preventive Medicine*, 36(1), 74-93. <https://doi.org/10.1016/j.amepre.2008.09.025>
- Livingston, M. (2011). A longitudinal analysis of alcohol outlet density and domestic violence. *Addiction*, 106: 919-925. <https://doi.org/10.1111/j.1360-0443.2010.03333.x>
- Melendez, R., Clarke, P. J., Khan, A. M., Gomez-Lopez, I. N., Li, M., & Chenoweth, M. (2020). National Neighborhood Data Archive (NaNDA): Socioeconomic status and demographic characteristics of census tracts, United States, 2008-2017 (V1) [Data set]. Inter-university Consortium for Political and Social Research. <https://doi.org/10.3886/E120462V1>
- Rummo, P. E., Meyer, K. A., Boone-Heinonen, J., Jacobs, D. R., Jr, Kiefe, C. I., Lewis, C. E., Steffen, L. M., & Gordon-Larsen, P. (2015). Neighborhood availability of convenience stores and diet quality: findings from 20 years of follow-up in the coronary artery risk development in young adults study. *American Journal of Public Health*, 105(5), e65-e73. <https://doi.org/10.2105/AJPH.2014.302435>
- Spring, A. (2018). Short- and long-term impacts of neighborhood built environment on self-rated health of older adults. *The Gerontologist*, 58(1): 36-46. <https://doi.org/10.1093/geront/gnx119>
- United States Census Bureau (2010). TIGER/Line shapefiles, 2010 census tracts (2010 version) [Data set]. https://www2.census.gov/geo/tiger/TIGER2010/TRACT/2010/tl_2010_us_tract10.zip.
- United States Census Bureau (2017). North American Industry Classification System (2017 version) [Data set]. <https://www.census.gov/cgi-bin/sssd/naics/naicsrch?chart=2017>
- Walls & Associates (2017). National Establishment Time Series (NETS) database.
- Xin, J., Zhao, L., Wu, T., Li, Y., Xue, H., Xiao, Q., Wang, R., Xu, P., Visscher, T., Ma, X., & Jia, P. (2019). Association between access to convenience stores and childhood obesity: A systematic review. *Obesity Reviews* 2019: 1-25. <https://doi.org/10.1111/obr.12908>
- Zenk, S. N., Schulz, A. J., Israel, B. A., James, S. A., Bao, S., & Wilson, M. L. (2005). Neighborhood racial composition, neighborhood poverty, and the spatial accessibility of supermarkets in metropolitan Detroit. *American Journal of Public Health* 95(4), 660-667. <https://doi.org/10.2105/AJPH.2004.042150>